



ANALYSIS OF DETERMINANTS OF ACCESS TO EXTENSION SERVICES AMONG WOMEN FARMERS IN NORTHERN ZONE OF PLATEAU STATE, NIGERIA

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ABSTRACT

The study examined the determinants of accessing extension services among women farmers in the Northern Agricultural zone of Plateau state, Nigeria. A multi-stage random sampling technique was used in selecting 240 farmers. Collected data were analyzed using both descriptive and inferential statistics. The result revealed that respondents had a mean age of 39 years, 86.7% are married with an average household size of 7 persons. It further revealed 73.3% of the farmers had formal education, while 77.1% of the respondents had no access to credit. More than half (50.4%) of the farmers had contact with extension agents and 62.9% of them did not belong to any farmer's association. Also, 54.2% of the respondents had annual average income of ₦ 322,783.33. Sources of agricultural information available to farmers were radio, friends, family, television, farmers and extension workers which ranked lowest. Logit regression's analysis result revealed that household size and level of education were the determinants influencing women farmers' access to extension services at ($p=0.05$) and ($p<0.10$), while access to credit was negatively significant at ($p<0.01$) respectively. Major constraints to accessing extension services were inadequate extension agents and land ownership. The study concluded that access to extension services, household size and level of education were the factors influencing extension services access among women farmers and with extension workers being the lowest ranked source of information, it was therefore recommended that more extension staff should be employed by the government and non-governmental organizations. More sensitization and tailored extension services should be delivered.

INTRODUCTION

Extension services refers to dissemination of useful and potential information relating to agriculture, practical application of important knowledge to farm and home, improving the livelihood in rural areas within the framework of the national, economical social politics involving the population as a whole (Jwanya *et al.*, 2014). The services attempt to provide the vehicle for increasing agricultural productivity because it links the farmers with proven agricultural innovation by exposing the rural farmers to ideas, information, improved farming methods and technology to enhance their productivity. It also provides linkages to veritable source of farm credits, markets as well as relevant training for value addition, processing, utilization of agricultural products. (In spite of the above and numerous side benefits provided through agricultural extension services, women farmers in rural communities in Nigeria hardly received adequate attention in agricultural extension coverage. Women play prominent roles in agricultural production and development, they

are deeply involved in the homestead production and the household, and this role cut across homemaking, childcare, family nutrition and other domestic chores. In the course of the above-mentioned activities, it is basically true that women are saddled with ensuring food sustainability in household or farm family, thus their involvement in agricultural production (Smith, 2010).

Agricultural technology delivery, which is the pillar of agricultural extension programme in Nigeria, as in many developing countries, is on the brink of collapse. Out of the six extension programmes that operated in Nigeria in the last thirty years, only two, River Basin Development Authority and the Agricultural Development Programme still exist and are being operated in limited scopes. The weak level of technology component of agriculture in Nigeria is manifested in the dismal performance of the agricultural sector and deepening poverty. Report from the Central Bank of Nigeria and the National Population Commission indicates that while the Nigerian population has been growing at 3% per annum, food production has been increasing at only 1.5% per annum in the last five years. In addition, 70% of Nigerians are living below poverty line (United Nations, 2013). However, women face numerous challenges in their access to agricultural extension services. These range from poor extension service delivery to marginalization of women farmers, insufficient funds, and limited involvement of women farmers in extension programmes. Limited coverage of extension services across rural areas, land tenure system (Iwena, 2015). Previous studies have shown that there is little or no literature that assesses extension service delivery among women crop farmers in the study area. It is against this backdrop this research was conducted.

Objective of the Study

The broad objective of the study is to analyze the extension service delivery to women farmers in the Northern Agricultural Zone of Plateau State, while the specific objectives were to describe the socio-economic characteristics of women crop farmers in the study area, identify the types of technology available to the respondents, describe different sources through which respondents receive extension information, determine the factors influencing women access to extension services and lastly describe the constraints to extension service delivery in the study area.

METHODOLOGY

i. The Study Area

Plateau State is located between latitude 8°24'N and longitude 8°32' and 10°38' east, with an area of 26,899 square kilometres. The State has an estimated population of about 3 million people (Plateau ICT Development Agency, 2021). The altitude ranges from around 1,200 meters (about 4000 feet) to a peak of 1,829 metres above sea level in the Shere Hills range near Jos. Years of tin mining have also left the area strewn with deep gorges and lakes (Plateau Diary, 2018). The two Local Governments Areas covered in Plateau State Northern Zone were (Jos East which lies between latitude 9°6'' and 29''E with total land area of 1,020km² and a population of 95,605 (Census, 2006), while Jos South lies between latitude 9°45'' and 54.22'' N and 8°45'N and 51°21' with a land area of 510km² and a population of 306,716 (Census, 2006). Though situated in the tropical zone, a higher altitude

means that Plateau State has a near temperate climate with an average temperature of between 18 and 22°C. Harmattan winds cause the coldest weather between December and February. The warmest temperatures usually occur in the dry season months of March and April. The mean annual rainfall varies from 131.75 cm (52 in) in the southern part to 146 cm (57 in) on the Plateau. The highest rainfall is recorded during the wet season months of July and August. The average lower temperatures have led to a reduced incidence of some tropical diseases such as malaria. The Jos Plateau, makes it the source of many rivers in northern Nigeria including the Kaduna, Gongola, Hadejia and Yobe rivers. The state is bounded by Bauchi to the North North, Kaduna to the Northwest, Taraba on the Northeast and Nassarawa state North South (Wuyep *et al.*, 2020).

The primary occupations of the people are farming, blacksmithery, marketing, civil service, artisan among others. The major crops grown by the people are cereals (acha, sorghum, maize, rice), root tuber (potato, carrot, sweet potato, cassava, cocoyam, yam) tree crops (avocado pear, olive, guava, pawpaw, orange, mango) and vegetables such as tomatoes, cabbage, lettuce, radish, passion fruits, pepper, spinach, cucumber, water melon just to mention few. The main livestock reared by the farmers are sheep, goats, pigs, rabbits, poultry and cattle. The main town is Jos city (Zembe, 2013).

ii. Sampling Procedure and Sample Size

A multi-stage sampling technique was used in selecting the respondents from the study area. In the first stage, Northern Zone of Plateau State ADP was purposively selected. In the second stage, two local governments were randomly selected from the six local governments. In the third stage three districts were randomly selected from the selected local governments. In the fourth stage 40 respondents were randomly selected making a total of 240 respondents randomly drawn from the study area.

iii. Methods of Data Collection

Primary data was obtained through the administration of structured questionnaire to the respondents. The questionnaire was categorized into sections base on the stated objectives.

iv. Method of Data Analysis

Data were analyzed using both descriptive and inferential statistics. The descriptive statistics used were means, frequency and percentages to analyze objectives i, ii, iii and (v) while inferential statistics used was logit regression analysis was used to analyze objective (iv).

v. Logit regression model

Binary logit model was used to analyse objective iv (Greene & Hensher, 2009). The adopted model was based on the structural specification in equation (1):

$$Y_i^* = \alpha + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + \epsilon_i \quad (1)$$

Where Y_i = latent (unobserved) extension service delivery.

X_i = Vector of explanatory variables, β_i = unknown regression parameters to be estimated, and ϵ_i = error effect. Latent variable Y_i , measured in ordinal scale is determined from the model.

The ordered logistic model which explains the factors influencing women crop farmers to extension service delivery is empirically specified in Eg. (i):

$$Y = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + b_4 X_4 + b_5 X_5 + \dots + E - 1$$

Where:

Y = Response of farmers to extension service delivery which is a dichotomous response variable (satisfied = 1, otherwise = 0).

X₁ = Age (Years)

X₂ = Marital status (single, divorced, widowed or married)

X₃ = Education (Number of years spent in formal education)

X₄ = Household size (Number of persons)

X₅ = Farming experience (Years)

X₆ = Membership of association (Yes = 1, No = 0)

X₇ = Extension Contact (Number of times)

X₈ = Farm size (Hectares)

X₉ = Access to credit (Access = 1, non-access = 0)

E = Error Term

Bi = Regression Coefficients

RESULTS AND DISCUSSIONS

i. Socio-economic Characteristics of the Respondents

Table 1 reveals the socioeconomic characteristics of the respondents. Majority (51.25%) of the respondents were within the ages of 31-50 years, while 27.5% and 21.55% of the respondents were within the ages of 11-30 and 51-70 years respectively. The mean age of the respondents was 39 years this implies that most of the respondents were within their active ages of production in agriculture. This implies that they were young and in their productive age. These findings were aligned with Vihi. *et al.* (2020) shows that more than half of the farmers were within their active age. The result further shows that more than half (86.7%) of the respondents were married, while only 13.3% of the respondents were single in the study area. This implies that most of the respondents were married women with dependent persons under their care (children) this is an attributing factor that will influence their participation in economic activities in order to cater for the needs of their families. This finding agreed with that of Obasi *et al.*, (2012) and Orisakwe and Agomuo (2011) who revealed in their separate studies that majority of farmers in Nigeria were married. The result also reveals that majority (88.75%) of the respondents had a household size of 1-10 persons. The mean average of 7 persons per household implies that most of the respondents had responsibilities to feed and care for their houses. The results agreed with that of Sennuga *et al.*, (2020) they posited that farmers in the North have large household size. This in line with findings of Orisakwe and Agomuo (2011) who asserted that large household is advantageous in farming as labour may be derived from the members. On the other hand the result indicates that majority (73.3%) of the respondents had attained tertiary education level, while 15% and 11.7% of the respondents had attained secondary and primary levels of education respectively. This implies that most of the respondents had attained one level of education or the other which is an important factor in adopting technologies. As reported an educated farmer is expected to possess good decision-making ability and thus, is able to take steps in adoption of innovations (Ozor & Njai, 2010). Kalungu and Filho (2016), asserted that educated farmers tend to adopt relevant agricultural technologies better than their illiterate counterparts.

Table 1: Distribution of women crop farmers according to Socio-Economic Characteristics

Variables	Frequency	Percentage	Mean
Age (Years)			
11-30	66	27.50	39.97
31-50	123	51.25	
51-70	51	21.55	
Marital Status			
Married	208	86.7	
Single	27	13.3	
Household Size			
1-10	213	88.75	7.0
11-20	27	11.25	
Education			
Primary	28	11.7	
Secondary	36	15.0	
Tertiary	176	73.3	
Access to credit			
Yes	55	22.9	
No	185	77.1	
Contact with Extension			
Yes	121	50.4	
No	119	49.6	
Membership of Association			
Yes	89	37.1	
No	151	62.9	
Years of Farming Experience			
1-20	172	71.67	15.77
21-40	56	23.33	
41-60	12	5.00	
Income (₦)			
100,000-200,000	94	39.2	322,783.33
201,000-300,000	36	15.0	
301,000-400,000	21	8.8	
401,000-500,000	55	22.9	
501,000 and Above	34	14.2	

Source: Field survey 2021

The result further shows that majority (77.1%) of the respondents had no access to credit, while 22.9% of the respondents had access to credit. This implies that most of the respondents had no access to credit which could hinder expansion of farms and adoption of technologies. The findings agreed with that of Kassem *et al.*, (2021) and Mallam (2015) who discovered in their separate studies that majority of rural farmers had no access to credit. The findings further reveals that half 50.4% of the respondents had extension contact, while 49.6% of the respondents had no extension contact. This implies that there is inadequate extension agents in the study area which could hinder technology transfer and adoption among women crop farmers. Orisakwe and Agomuo (2011) noted that regular contact with extension agents motivates and exposes the farmers to innovations and gives them information on how to use the technologies. The result also indicates that majority 62.9% of the respondents were not members of any farmers association, 37.1% of the

respondents indicates membership of farmers association. The result also shows that majority (71.67%) of the respondents had 1-20 years of farming experience, with an average of 15 years of farming experience among women crop farmers it shows that most of them had attained certain experience in farming and will be willing to accept and adopt new technologies. The findings supports that of Okoye (2006) who stated that more experienced farmers are expected to have higher levels of technical efficiency. The result further shows that respondents had an average annual income of ₦322,783.33 which implies that women crop farmers were not peasant farmers and had good annual income.

Technology Available to Women Crop Farmers

The result from Table 2 shows that the respondents received messages on varietal technology, pest control and weed control which were ranked 1st and 3rd respectively. They are followed by harvesting, storage, disease control and land conservation technologies which were ranked 4th, 5th, 6th and 7th respectively. Other were spacing, home management and land preparation, planting technologies which were ranked 8th, 10th and 11th respectively. This implies that majority of the respondents had access to one form of technologies or the other. The above results corroborate the findings of Nasiru *et al.*, (2002) which reveals that the crop sub-sector still dominates both in number of technologies pushed to farmers and their level of adoption. However, it is contrary to the findings of Idrisa and Ogunbameru (2008) which portrays livestock sub-sector dominating technologies pushed to farmers.

Table 2: Distribution of Respondents Based on Technology Available to them

Technology Available	Percentage	Rank
Varietal Technology	90.00	1 st
Pest Control	90.00	1 st
Weed Control	85.40	3 rd
Harvesting	84.20	4 th
Storage	82.50	5 th
Disease Control	80.80	6 th
Land Conservation	80.00	7 th
Spacing	79.60	8 th
Home Management	79.60	8 th
Land Preparation	76.70	10 th
Planting	75.40	11 th
Total	905*	

Source: Field Survey 2021

* Multiple responses

Sources of Extension Information to Women Crop Farmers

The result from Table 3 shows that the respondents had different sources of extension information most of the respondents indicated that radio, friends/relations and Television were the major sources of information and were ranked 1st, 2nd and 3rd respectively. The results also shows that farmer to farmer, field days, result demonstrations, method demonstrations and individual contact with extension agent were ranked 4th, 5th, 6th, 7th and 8th respectively as a source of information to the women crop farmers. This implies that most of the respondents had different sources of information on the different types of technologies. This is in line with Sennuga, *et al.*, (2020) that farmers preferred traditional ICT, mainly radio as their main source of accessing agricultural information. The above findings are further corroborated by Muktar *et al.*, (2016) who affirmed that only about

46.3% farmers are aware that there is still extension services being giving as most farming and market information are sourced either through a friend or through radio.

Table 3: Distribution of Respondents Based on Sources of Information

Source of Extension of Information	Percentage	Rank
Radio	90.00	1 st
Friends & Relations	89.60	2 nd
Television	87.90	3 rd
Farmer to farmer	79.20	4 th
Field days	74.60	5 th
Result demonstration	70.00	6 th
Method demonstration	63.30	7 th
Individual contact with extension agent	37.90	8 th
Total	592.5*	

Source: Field Survey 2021

*Multiple Response

Factors that Influence Women Crop Farmers Access to Extension Services

The result from Table 4 shows the factors influencing extension service delivery among the respondents. The findings reveal that access to credit and household size were the major factor influencing extension service delivery among the respondents in the study area which was significant at 5% ($P>0.05$). Household size was positive which implies that as household size increases the extension service delivery also increases. On the other hand, access to credit was found to be negative and significant in influencing extension service delivery among the women crop farmers, which means the access to credit decreases and extension service delivery also decreases among the women farmers. This implies that household size and access to credits highly determines level extension service accessibility among women farmers that is the higher the number of persons per household the more the access to extension delivery services and the higher the accessibility to credits the better access to extension agents' services delivery. This is in agreement with Silong, A. And Gadanaki (2020) who reported that household Size influences the probability of credit access positively. On the other hand, educational level of the respondents also influence access to extension service delivery and was significant at 10%, while age, marital status, income, years of farming experience, position in the household, extension contact and membership of association were not in any way influencing extension service delivery in the study area which shows that the few factors will be improved so as to improve extension service delivery in the study area.

Table 4: Logistic regression estimates of the factors that influences women crop farmers access to extension service delivery

Variables	Coef.	Std. Err.	z	P> z
Age	.0013558	.0218981	0.06	0.951
Marital Status	1.021281	.7675145	1.33	0.183
Household Size	.1096307	.0556612	1.97	0.049
Level of Education	.499507	.2664134	1.87	0.061
Income	-3.11e-07	1.85e-07	-1.68	0.092
Years of Farming Experience	-.0302585	.0222152	-1.36	0.173
Access to Credit	-2.724348	.8828985	-3.09	0.002
Positioning the Household	-.6065399	.4975144	-1.22	0.223
Contact with Extension Agent	.290963	.4334474	0.67	0.502
Membership of Association	-.2561677	.4933766	-0.52	0.604
Constant	5.167306	2.295018	2.25	0.024

Source: Field Survey 2021

Constraints to Women crop Farmer's Access to Extension Service Delivery

The result from Table 5 shows the constraints faced by respondents on extension service accessibility in the study area where inadequate extension agents, land ownership and conflict were the major constraints faced by the respondents and were ranked 1st, 2nd and 3rd respectively, while inadequate supervision, household activities, cultural barriers, inadequate advice from women extension agents, women crop farmers misinformed, neglect of extension workers advice and inadequate credit were ranked 4th, 5th, 6th, 7th, 8th, 9th and 10th respectively.

Table 5: Constraints to Women crop Farmer's Access to Extension Services

Contacts	Percentage*	Rank
Inadequate of Extension Agents	76.20	1 st
Land ownership	73.80	2 nd
Conflict	71.70	3 rd
Inadequate Supervision	71.20	4 th
Household Activities	67.50	5 th
Cultural Barriers	65.40	6 th
Lack of Advice from Women Extension Agents	64.20	7 th
Women Crop Farmers Misinformed	61.20	8 th
Neglect of Extension-Worker's advice	58.30	9 th
Lack of Credit	54.20	10 th
Total	663.7	

Source: Field Survey 2021

*Multiple Response

CONCLUSION

Based on the findings, it can be concluded that access to credit and household size were the major factors influencing extension service delivery among women crop farmers in the study area. Therefore, the need to women farmer access to credit cannot be overemphasized because it has potentials by implication improving their productivity and welfare.

RECOMMENDATIONS

Based on the above the following recommendations were made:

- i. Government should make more effort to employ more extension agents to bridge the gap between farmers and extension agents.
- ii. Women crop farmers should be supported with funds by the Federal government to acquire sufficient land.
- iii. Government should find a way of bringing peace between conflicting farmers and herders.
- iv. Women should be advised to prioritize extension meetings in their programmes by suspending every household activity to attend.
- v. Female extension agents should be assigned to deliver extension message to women farmers in places where there are cultural barriers.

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